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Systems and methods of generating a design based on a user search query

Abstract

An apparatus includes network interface and a processor. The network interface is configured to receive first user input indicating a first search query. The processor is configured to identify one or more first keywords by parsing the first search query, to select a first design template based on the one or more first keywords, and to select first design content based on the one or more first keywords. The first design content includes at least one of a first font, a first fill, or a first image. The processor is also configured to generate a first design based on the first design template and to update the first design based on the first design content. The processor is configured to generate a first graphical user interface (GUI) including a first design image of the first design and to provide the first GUI via the network interface to a client device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of the filing date of International PCT Patent Application No. PCT/AU2020/050651, filed on Jun. 26, 2020, that claims priority to U.S. Provisional Patent Application No. 62/868,104, filed on Jun. 28, 2019, both of which are hereby incorporated by reference in their entirety.

BACKGROUND

(1) With the proliferation of design tools, there is increasing use of templates to generate designs. Typically, a user selects categories or provides keywords to search for pre-existing templates. The user selects a template from the search results and adds user content. The pre-existing templates are generally based on common design practices and may not closely match the keywords provided by the user. Providing the keywords initially to search for a template and then adding user content to more closely match the initially provided keywords is repetitive, time-consuming, and requires user effort to search for content.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a particular example of a system that is operable to generate a design based on a user search query;
- (2) FIG. 2 illustrates a particular example of a user interface that may be generated by the system of FIG. 1;
- (3) FIG. 3 illustrates a particular example of a search query received via a user interface of the system of FIG. 1;
- (4) FIG. 4 illustrates particular examples of a search query and a corresponding design that may be generated by the system of FIG. 1;
- (5) FIG. 5 illustrates particular examples of a search query and a corresponding design that may be generated by the system of FIG. 1;
- (6) FIG. 6 illustrates a particular example of a search query and a corresponding design that may be generated by the system of FIG. 1;
- (7) FIG. 7 illustrates a particular example of a search query and corresponding designs that may be generated by the system of FIG. 1;
- (8) FIG. 8 illustrates a particular example of a search query and a corresponding design that may be generated by the system of FIG. 1;
- (9) FIG. 9 illustrates a particular example of a user interface to edit a design that may be generated by the system of FIG. 1;
- (10) FIG. 10 illustrates another particular example of a system that is operable to generate a design based on a user search query; and
- (11) FIG. 11 illustrates a particular example of a method of generating a design based on a user search query.

DETAILED DESCRIPTION

(12) Design templates are a popular way to generate graphic designs, e.g., for websites, brochures, invitations, etc. In accordance with particular embodiments of the present disclosure, a user provides a search query by typing into a search field. The search query indicates what the user would like to create (e.g., “a red invitation”). One or more keywords (e.g., “red”, “invitation”, and

“red invitation”) are identified by parsing the search query (e.g., “a red invitation”). A design template (e.g., an invitation template) is selected based on the keywords. As used herein, a “template” includes a default layout of images, text, colors, and/or fonts. For example, a placeholder template includes text elements that are placed on a white background. In a particular aspect, the layout of the text elements is based on a type of the template. For example, an Instagram® (a registered trademark of Instagram LLC, Menlo Park, CA) post template includes a pre- title text element that includes default pre-title text (e.g., “Tap to add a pre-title (optional)”), a title text element that includes default title text (e.g., “Tap to add text”), and a subtitle text element that includes default subtitle text (e.g., “Tap to add a subtitle (optional)”).

(13) The design template (e.g., an invitation template) is used to generate a design. As used herein, a “design” includes one or more design elements and format information (e.g., color information, font type, font size, etc.). A “design element” includes image content (e.g., in vector format or pixel format), text content, fill content, etc. In a particular aspect, the format information includes first format information corresponding to a first design element type (e.g., a title text element). For example, the first format information (e.g., default format information) indicates color information, font type, font size, etc. associated with design elements of the first design element type. In a particular aspect, the first format information indicates default format information that is used to format a first design element of the first design element type that is added to the design. Thus, the format information may cause a “title” design element to have a different font face and size than other types of design elements.

(14) In a particular aspect, the design elements include one or more attributes. For example, the first design element includes second format information, one or more additional attributes, or a combination thereof. In a particular aspect, a user updates a format of the first design element and the second format information indicates the updated format of the first design element. In a particular aspect, the second format information overrides the first format information (e.g., the default format information) for the first design element. For example, the first design element is formatted based on the second format information. To illustrate, an invitation template may have red title text by default and the user may change the title text of a first title element of an invitation to blue. If the user adds a second title element to the invitation, the second title element has red title text by default (as indicated by the first format information) while the first title element keeps the blue title text (as indicated by the second format information of the first title element).

(15) Generating the design based on the design template includes adding second design elements to the design that correspond to first design elements of the design template. For example, an image element is added to the design based on determining that the design template includes an image element. An image (e.g., a cupcake image) from the image element of the design template is copied to the image element of the design.

(16) Design content is selected based on the keywords (e.g., “red”, “invitation”, and “red invitation”) and the design is updated based on the selected design content. For example, a red fill element is selected based on the keywords (e.g., “red” and “red invitation”). The design is updated to have a red fill element as a background element. A design image of the design may be provided to a display. The user can select the design image to view or edit the design.

(17) Updating the design based on the keywords enables the design to be customized based on the search query without the user having to provide additional input. For example, the design is updated automatically based on the search query prior to presenting a design to the user and prior to the user having a chance to modify a presented design. The user is saved time and effort by having the design automatically modified to more closely match what the user is planning to do. The ability to customize the design based on the keywords expands the available design options. As an illustrative example, three design templates including one image each can be used to automatically generate at least nine designs. For example, a first design, a second design, and a third design can be generated by adding a first image, a second image, and a third image,

respectively, to a first design template. Additional illustrative, non-limiting advantages of the present disclosure are described herein.

(18) In the following description, certain aspects of the present disclosure are described in conjunction with a graphic design website that enables users to create designs using web application(s) or other Software-as-a-service (SaaS) components. Examples of graphic designs include, but are not limited to, social media posts or headers, presentations, letterheads, magazine templates, certificates, resumes, flyers, book/album covers or art, infographics, flyers, posters, menus, business cards, brochures, logos, greeting cards, postcards, invitations, collages, announcements, blog graphics, Internet advertisements, videos, animations, etc. In some examples, the graphic design website can be accessed by using Internet browsers and/or an application installed on a mobile device, such as a mobile phone or tablet. Unless specified otherwise, reference to the “graphic design website” herein includes the graphic design web application(s)/SaaS component(s) accessible via the graphic design website. Use of the graphic design website to create designs may be faster and/or more convenient than using a standalone downloaded/purchased software program. However, it is to be understood that the description of website-based aspects herein is for illustration only and is not to be considered limiting. In alternative implementations, the techniques of the present disclosure may be applied in the context of a standalone computer application executed on a desktop computer, a laptop computer, a mobile device, etc. A “user” can refer to an internal user or an external user of the graphic design website. For example, an internal user includes a designer, an employee, a software engineer, or another person that manages or designs the graphic design website. An external user includes a person that is not an internal user, such as someone that accesses the graphic design website to generate designs and does not manage or design the graphic design website. In a particular aspect, one or more operations described herein with reference to a user can be performed by a first user (e.g., an internal user) and one or more operations described herein with reference to the user can be performed by a second user (e.g., an external user).

(19) When a user provides a search query (e.g., “a holiday card with a puppy”) on the graphic design website, a new design (e.g., a first design) is generated based on a blank design (e.g., a placeholder template) or a design template that has a pre-applied layout. For example, the graphic design website may have access to several “holiday greeting card” templates, and may select one of the templates as a starting point from which to customize a holiday greeting card based on keywords of the search query. In a particular example, a design template (e.g., a holiday card template) is selected based on one or more of the keywords (e.g., “holiday card”) of the search query and a first design (e.g., a holiday card design) is generated based on the design template. Design content (e.g., an image of a puppy) is selected based on one or more of the keywords (e.g., “puppy”) of the search query. The first design (e.g., the holiday card design) is updated based on the design content. For example, the first design includes an image (e.g., a default image) copied from the design template (e.g., the holiday card template). The first design is updated to replace the default image with the selected image (e.g., the puppy image).

(20) The user may edit (e.g., further customize) the first design by uploading photos of their own family, entering a personalized greeting message, changing the images/text/fonts/colors, etc. To provide a more comprehensive design experience, the graphic design website may enable the user to search for images, templates, fonts, etc., as described above. To illustrate, the user may enter the search term “dog” to search for stock images of dogs that can be inserted into a design.

Incorporating a search result image (e.g., a photo of a dog) into a design may be as simple as “dragging-and-dropping” the image into the design, at which point the user may be free to edit the image in terms of size, color, etc. Certain media assets (e.g., images, templates, fonts, etc.) may be free to incorporate into designs while others may be available for a price and may have an associated license (e.g., a one-time-use license) that is enforced by the graphic design website.

(21) The graphic design website generates a graphical user interface (GUI) including a first design

image of the first design. The graphic design website provides the GUI to a display. In a particular aspect, the user can select the first design image to edit the first design. For example, the graphic design websites update the GUI to include one or more options for editing the first design, an option to save the design, one or more options to publish the design, or a combination thereof.

(22) An illustrative example of a system that supports operation of such a graphic design website and that is operable to generate a design based on a user search query is shown in FIG. 1 and is generally designated **100**. The system **100** includes a client device **192** that is communicably coupled, via a content delivery network (CDN) **101**, to a computing environment (e.g., a network-accessible cloud computing environment). For example, the client device **192** is communicatively coupled to one or more design service(s) **126** of the computing environment. In a particular aspect, the design service(s) **126** correspond to one or more services or components described with reference to FIG. 10, one or more additional services or components, or a combination thereof. Although FIG. 1 shows a CDN **101**, it is to be understood that in alternative aspects devices shown in FIG. 1 may communicate using one or more networks (or portions thereof) that include but are not limited to CDN(s), private network(s), cellular network(s), the Internet, etc. The client device **192** is configured to be operated by a user **104**. As further described herein, the system **100** may enable the user **104** to quickly and easily create and edit various forms of graphic designs and other visual media works, even if the user **104** is not a trained graphic designer or other visual arts professional. In some examples, the computing environment includes one or more servers and/or one or more databases, data stores, and/or data storage devices. For example, the computing environment includes one or more design database(s) **184**. In a particular aspect, the design database(s) **184** correspond to one or more databases or components described with reference to FIG. 10, one or more additional databases or components, or a combination thereof. The computing environment may include one or more load balancers, as further described with reference to FIG. 10.

(23) In particular aspects, the client device **192** corresponds to a desktop computer, a laptop computer, a mobile phone, a tablet, or other computing device configured to access websites via the Internet. The client device **192** may include (or may be connected to) one or more input devices (e.g., a keyboard, a mouse, a touchscreen, etc.) and one or more output devices (e.g., a monitor, a touchscreen, audio speakers, a headphone connection, etc.).

(24) In particular aspects, the client device **192**, the computing environment, the one or more servers, the design database(s) **184**, the load balancers, or a combination thereof, as further described with reference to FIG. 10, may include one or more processors or processing logic, memories, and/or network interfaces. The memories may include instructions executable by the processors to perform various functions as described herein. The network interfaces may include wired and/or wireless interfaces operable to enable communication to local area networks (LANs) and/or wide area networks (WANs), such as the Internet.

(25) A first server may be configured to execute a web front end. The web front end is configured to serve a graphic design website to requestors. For example, the web front end may send one or more graphical user interfaces to the client device **192** in response to receiving a request from the client device **192**. The same or a different (e.g., second) server may be configured to execute a design service(s) **126**. The design service(s) **126** may enable core design creation, design updating, and design deletion functionality of the graphic design website. The same or a different (e.g., third) server may be configured to execute an export service. The export service may support graphic design export tasks (e.g., publishing a design as a web output).

(26) The load balancers may be configured to send a message to a server requesting instantiation of a particular service. The server may execute a virtual machine including an instance of the particular service. For example, the load balancers may be configured to send a message to the second server to instantiate the design service(s) **126**. The second server may, in response to receiving the message, execute a virtual machine including an instance of the design service(s) **126**.

In a particular aspect, multiple servers may execute instances of the same service. For example, the load balancers may send a second message to a fourth server to instantiate the design service(s) **126**. In this example, the second server may execute a first virtual machine executing a first instance of the design service(s) **126** and the fourth server may execute a second virtual machine including a second instance of the design service(s) **126**.

(27) The design service(s) **126**, the export service, or both, may be coupled to one or more endpoints. In FIG. **1**, the design service(s) **126**, the export service, or both may be coupled, via the CDN **101**, to an endpoint **172**, an endpoint **174**, one or more additional endpoints, or a combination thereof. An “endpoint” may also be included in the client device **192**. For example, the client device **192** may include the endpoint **172**. An endpoint may refer to an application, a server, a social network site, a web server, a device, or a combination thereof.

(28) During operation, the user **104** may operate the client device **192** to access a graphic design website via the computing environment. The graphic design website may enable the user to create and modify a design **103** (which may be persisted to a document database, as further described with reference to FIG. **10**) using web application(s) and/or Software-as-a-service components presented by the graphic design website. For example, the web front end may generate a graphical user interface (GUI) **181** of the graphic design website. The user **104** may use the GUI **181** to create and modify the design **103**. Turning to the example illustrated in FIG. **2**, the GUI **181** includes a search field **202** (e.g., a text field) and a background **204**. In a particular aspect, the background **204** includes a default background image. The user **104** provides the user input **109** by typing in the search field **202**. It should be understood that typing is provided as illustrative example of providing the user input **109**. In other examples, the user **104** provides the user input **109** in other ways, including but not limited to, audio input, video input, image input, other type of sensor input, or a combination thereof. In a particular example, the user **104** provides the user input **109** by speaking into a microphone of a mobile device. The mobile device includes a speech-to-text engine that generates text based on an audio signal and updates the search field **202** based on the text.

(29) In the example illustrated in FIG. **3**, the user **104** enters (e.g., by typing, speaking, movement, etc.) first text (e.g., “A red”) in the search field **202**. In a particular example, the design service(s) **126** processes the user input **109** as each character or word is entered in the search field **202**. In an alternative example, the GUI **181** includes a send option (not illustrated) and the design service(s) **126** processes the user input **109** of the search field **202** in response to receiving a selection of the send option. The user input **109** indicates a search query **117** (e.g., “A red”). The design service(s) **126** generates one or more keywords **127** (e.g., “red”) by parsing the search query **117**. The design service(s) **126** determines whether the keywords **127** indicate a design template type (e.g., a greeting card, an invitation, an Instagram® post, etc.). The design service(s) **126** refrains from generating the design **103** in response to determining that the keywords **127** (e.g., “red”) do not indicate any design template type.

(30) In a particular aspect, the design service(s) **126** updates the background **204** in response to determining that the keywords **127** (e.g., “red”) indicate a color. For example, the design service(s) **126** updates the background **204** to have the color (e.g., “red”) indicated by the keywords **127**. To illustrate, the design service(s) **126** updates the background **204** by applying a first filter (e.g., a red filter) to the background **204**. Updating the background **204** based on the keywords **127** (e.g., “red”) may signal to the user **104** that the design service(s) **126** is actively processing (e.g., in real-time) the user input **109** provided by the user **104** via the search field **202**.

(31) The user **104** may continue to enter input (e.g., type) in the search field **202**. In the example illustrated in FIG. **4**, the search field **202** indicates an updated version of the search query **117** (e.g., “A red invitation”). The design service(s) **126** identifies the keywords **127** (e.g., “red”, “invitation”, and “red invitation”) by parsing the search query **117** (e.g., “A red invitation”). In a particular aspect, design template mapping data indicates keywords corresponding to each design template type. For example, the design template mapping data indicates that a first keyword (e.g.,

“invitation”) matches a particular design template type (e.g., an invitation design template type). The design service(s) **126**, in response to determining that the keywords **127** include the first keyword, determines that the keywords **127** indicate the particular design template type (e.g., an invitation design template). The design service(s) **126** selects a design template **105** from design templates **111** based on the keywords **127**. For example, the design service(s) **126** selects the design template **105** in response to determining that a type **167** of the design template **105** indicates the particular design template type (e.g., indicated by the first keyword).

(32) In a particular example, the design service(s) **126** retrieves tag data **145** from the design database(s) **184**. The tag data **145** includes one or more template tags associated with the design template **105**. For example, the one or more template tags include a cupcake image tag, a dessert image tag, a food image tag, a berry image tag, a frosting image tag, or a combination thereof, indicating a type of image included in the design template **105**. In a particular example, the one or more template tags include an event tag (e.g., a 16th birthday tag) indicating that the design template **105** is associated with a particular event (e.g., a 16th birthday). The one or more template tags include color tags (e.g., a red tag, a white tag, a brown tag, or a combination thereof) indicating colors of design elements included in the design template **105**. In a particular aspect, the color tags indicate colors that can be assigned to design elements of the design template **105**. For example, a color tag indicates that a background element of the design template **105** can have any color. As another example, a color tag (e.g., a bright-color family tag) indicates that a background element of the design template **105** can have any color in a color family (e.g., a bright-color family). For example, a bright-color family includes bright shades of red, blue, and green, and does not include pastel colors. In a particular example, the color tag (e.g., any-color-except-blue) indicates that a background element of the design template **105** can have any color except a particular color that is included in a foreground element (e.g., blue text). In a particular aspect, the design service(s) **126** selects the design template **105** based at least in part on determining that at least one of the template tags (e.g., a red tag, an any-color tag, a bright-color family tag, or an any-color-except-blue tag) of the design template **105** “matches” at least one of the keywords **127** (e.g., “red”). As used herein, “match” is not limited to an identical word to word correspondence. As an example, a “red” keyword can match a red tag, an any-color tag, a bright-color family tag, or an any-color-except-blue tag.

(33) In a particular aspect, the design service(s) **126** identifies the design templates **111** based on the keywords **127**. For example, one or more of the design templates **111** may have a design template type (e.g., a party template, a birthday template, an invitation template, an event template, or a card template) that matches one or more of the keywords **127** (e.g., “invitation”). The design service(s) **126** determines scores of the design templates **111** based on the keywords **127**. For example, the design service(s) **126** determines a first score of the design template **105** based on a count of the first template tags of the design template **105** that match at least one of the keywords **127**. In a particular aspect, the design service(s) **126** retrieves search selection data **107** from the design database(s) **184**. The search selection data **107** indicates that a second design was saved by a second user at a first time, and further indicates that the second design was based on a second search query and the design template **105**. In a particular aspect, the design service(s) **126** determines, at a second time, the first score based at least in part on a comparison of the first time and the second time, a comparison of the search query **117** and the second search query, a comparison of the first user characteristics of the user **104** (e.g., a logged-in user) and second user characteristics of the second user, or a combination thereof. In a particular example, the first score increases with the recency that the second design based on the same design template was saved (e.g., a difference between the first time and the second time is less than a threshold). In a particular example, the first score is higher if the same design template was used to generate the second design responsive to a common search query terms and the second design was saved by a user (e.g., a count of common keywords associated with the search query **117** and the second search query is

greater than a threshold). In a particular example, the first score is higher if the second design is based on the same design template and if the second design was saved by a user having common characteristics as the user **104** (e.g., a count of common characteristics associated with the user **104** and the second user is greater than a threshold). In a particular aspect, the first score increases based on a count of the common characteristics. User characteristics may include age, location, gender, occupation, ethnicity, hobbies, interests, social group membership, music preferences, reading preferences, web browsing preferences, location history, food preferences, or a combination thereof. As an illustrative example, the first score is calculated in manner that increases a likelihood of selecting a particular design template that is recently popular among users having similar characteristics as the user **104**. In a particular aspect, the design service(s) **126** selects the design template **105** based on determining that the first score satisfies (e.g., is greater than) a score threshold (e.g., 0), that the first score is highest among the scores of the design templates **111**, or both. In a particular aspect, the design service(s) **126** refrains from using personally identifiable information of other users to select design templates for the user **104**. In a particular aspect, the design service(s) **126** may select design templates for the user **104** based on personally identifiable information of other users only with explicit consent of the other users.

(34) The design service(s) **126** generates the design **103** based on the design template **105**. For example, the design template **105** includes one or more design element(s) **141**, format information **165**, the type **167**, or a combination thereof. The design service(s) **126** generates the design **103** including one or more design element(s) **121** corresponding to the design element(s) **141**. For example, the design service(s) **126** adds a design element **123** to the design **103** in response to determining that the design template **105** includes a design element **143**. The design service(s) **126** copies one or more attributes of the design element **143** to the design element **123**.

(35) In a particular example, the one or more attributes of the design element **123** (e.g., copied from the design element **143**) include a type **171**, a content role **179**, text information **173**, a media identifier **189**, fill information **177**, bounding block information **185**, a position **175**, a page tag **186**, or a combination thereof. The type **171** indicates whether the design element **123** is or includes an image element, a text element, or a fill element. In a particular aspect, the type **171** indicates whether the design element **123** is or includes a foreground element or a background element.

(36) The content role **179** of the design element **123** indicates a role of the design element **123** in the design **103**. Some illustrative examples of content roles include style-related content roles (e.g., title, pre-title, subtitle, heading, or paragraph) and information structure-related content roles (e.g., name, address, invoice-tax-details, or invoice-items-purchased). In a particular aspect, a first set of content roles is associated with a first design template type. For example, the first set of content roles associated with Instagram® post template type includes a pre-title content role, a title content role, and a subtitle content role. A second set of content roles associated with another design template type may include more, fewer, or different content roles. A content role enables identification of corresponding design elements in designs and design templates.

(37) When the design element **123** includes a text element, the text information **173** indicates text (if any) included in the design element **123**. In some examples, the text information **173** indicates format information (e.g., second format information) of the design element **123**. When the design element **123** includes an image element, the media identifier **189** indicates an image (if any) included in the design element **123**. When the design element **123** includes a fill element, the fill information **177** indicates a fill color of the design element **123**. The bounding block information **185** indicates a shape, a size, a height, a width, or a combination thereof, of a bounding block of the design element **123**. The page tag **186** indicates a design page (e.g., a first page, a second page, etc.) of the design **103** that includes the design element **123**. The position **175** indicates a location of the design element **123** on the design page of the design **103**.

(38) In a particular aspect, the design service(s) **126** initializes a type **157** (e.g., invitation design) of

the design **103** based on the type **167** (e.g., invitation template) of the design template **105**. In a particular aspect, the design service(s) **126** initializes format information **155** of the design **103** by copying the format information **165** of the design template **105**.

(39) The design service(s) **126** selects design content **119** based on the keywords **127** (e.g., “red”, “invitation”, “red invitation”, or a combination thereof in the examples of FIGS. 4-6). The design content **119** includes a fill **135**, an image **115**, a font **125**, or a combination thereof, that matches at least one of the keywords **127**. For example, the design service(s) **126**, in response to determining that at least one of the keywords **127** indicates a first color (e.g., “red”), sets the fill **135** to the first color.

(40) In a particular example, the design service(s) **126** performs natural language analysis techniques to determine that a second keyword (e.g., “red”) is a modifier of the first keyword (e.g., “invitation”). In some examples, the design service(s) **126**, in response to determining that the second keyword (indicating a color) is a modifier of the first keyword (indicating a particular design template type), sets the fill **135** to the color indicated by the second keyword.

(41) The design service(s) **126** updates the design **103** based on the design content **119**. For example, the design service(s) **126** updates the design **103** to include a fill element (e.g., the design element **123**) indicating the fill **135** of the first color. The fill element is added as a background element to the design **103**. In a particular aspect, the design service(s) **126** updates the fill information **177** of a background fill element of the design **103** to include the fill **135** (e.g., a red fill). In an alternative aspect, the design service(s) **126** replaces another background element of the design **103** with the fill element (e.g., the design element **123**). In a particular aspect, the design service(s) **126** adds the fill element (e.g., the design element **123**) as a background element to the design **103**.

(42) The design service(s) **126** update the GUI **181** to include a design image **129** (e.g., a thumbnail image) of the design **103**. In a particular aspect, the design service(s) **126** update the background **204** based on a color of the background element of the design **103**. For example, the background **204** is updated to have a second color that is distinct from a first color of the background element. Updating the background **204** to have a second color that is distinct from the first color enables the design image **129** to stand out to the user **104** relative to the background **204**. The design service(s) **126** provides the updated version of the GUI **181** to the client device **192**.

(43) The user **104** may continue to enter input (e.g., type) in the search field **202**. In the example illustrated in FIG. 5, the search field **202** indicates an updated version of the search query **117** (e.g., “A red invitation with a puppy”). The design service(s) **126** identifies the keywords **127** (e.g., “red”, “invitation”, “red invitation”, “puppy”) by parsing the search query **117**.

(44) In a particular aspect, the user **104** can update the user input **109** to incrementally update the design **103** by updating the design content **119** (e.g., without changing the design template). In a particular example, the design service(s) **126** selects the image **115** based on determining that a first tag (e.g., dog) associated with the image **115** matches a keyword (e.g., puppy) of the keywords **127**. In a particular aspect, the design service(s) **126** generates scores based on a count of tags of the images **113** that match at least one of the keywords **127**. In this aspect, the design service(s) **126** selects the image **115** from the images **113** based on a first score of the image **115** (e.g., greater than a threshold score, a highest score, or both). The design service(s) **126** adds or updates an image element (e.g., the design element **123**) of the design **103** to include the image **115**. For example, the media identifier **189** of the design element **123** indicates an identifier of the image **115**.

(45) The design service(s) **126** updates the GUI **181** to include an updated version of the design image **129**. The updated version of the design image **129** includes an image of the updated version of the design **103**. The design service(s) **126** provides the updated version of the GUI **181** to the client device **192**.

(46) The user **104** may continue to enter input (e.g., type) in the search field **202**. In the example illustrated in FIG. 6, the search field **202** indicates an updated version of the search query **117** (e.g.,

“A red invitation with a puppy that says “Happy barkday””). The design service(s) **126** identifies the keywords **127** (e.g., now including the quote “Happy barkday”) by parsing the search query **117**. In a particular aspect, the design service(s) **126** parses quoted text in the search query **117** as a text keyword. The design service(s) **126** updates the design **103** to include a text element (e.g., the design element **123**) that includes the text keyword. For example, the design service(s) **126** updates a text element of the design **103** to include the text keyword, replaces a first text element of the design **103** with a second text element that includes the text keyword, or adds a text element that includes the text keyword to the design **103**. In a particular aspect, the text element that indicates the text keyword has a particular content role (e.g., a title content role).

(47) In a particular aspect, the search query **117** can include text style information (e.g., “A red invitation with a puppy that says “Happy barkday” in Times New Roman”). Keywords (e.g., “Times New Roman”) may indicate a font **125** (e.g., Times New Roman font). The design service(s) **126** selects the font **125** (as part of the design content **119**) in response to determining that one or more of the keywords **127** indicate the font **125**. The design service(s) **126** updates the design **103** based on the font **125**. In a particular example, the design service(s) **126** updates the format information **155** to indicate the font **125** for each design element type. In another example, the design service(s) **126** updates the format information **155** to indicate the font **125** for a particular design element type (e.g., a title design element type). In a particular implementation, the design service(s) **126** determines that a text parsing technique (e.g., a parse tree) indicates that second keywords (e.g., “Times New Roman”) indicating the font **125** are modifiers of one or more first keywords (e.g., “Happy barkday”) indicating the text keyword. The design service(s) **126**, in response to determining that the second keywords (e.g., “Times New Roman”) indicating the font **125** are modifiers of the first keywords (e.g., “Happy barkday”) indicating the text keyword, updates the text information **173** of the text element (e.g., the design element **123**) to indicate the font **125**.

(48) The design service(s) **126** updates the GUI **181** to include an updated version of the design image **129**. The updated version of the design image **129** includes an image of the updated version of the design **103**. The design service(s) **126** provides the updated version of the GUI **181** to the client device **192**.

(49) In a particular implementation, the design service(s) **126** generates multiple designs based on the keywords **127**. For example, the design service(s) **126** selects a second design template based on the keywords **127**. In some examples, the second design template is distinct from the design template **105**. In a particular aspect, the second design template corresponds to a next highest score among the scores for the design templates **111**. The design service(s) **126** generates a second design based on the second design template. The design service(s) **126** select second design content based on the keywords **127**. In some examples, the second design content is distinct from the design content **119**. The design service(s) **126** updates the second design based on the second design content. The design service(s) **126** generates a design image of the second design. In the example illustrated in FIG. 7, the GUI **181** includes the design image **129** and a design image **729** of the second design. Including multiple design images on the GUI **181** enables comparison of the design **103** and the second design. In a particular aspect, the user **104** selects the design image **129** or the design image **729** to edit the corresponding design, as further described with reference to FIG. 9.

(50) The user **104** may continue to enter input (e.g., type) in the search field **202**. In the example illustrated in FIG. 8, the search field **202** indicates an updated version of the search query **117** (e.g., “A red invitation with a puppy that says “Happy barkday” and balloons”). The design service(s) **126** identifies the keywords **127** (e.g., now including the keyword “balloons”) by parsing the search query **117**. In a particular aspect, the design service(s) **126** updates a design template based on updated keywords. For example, the design service(s) **126** selects a second design template based on the keywords **127** (e.g., “red”, “invitation”, “red invitation”, “puppy”, “balloons”). In a particular aspect, the GUI **181** includes a reset option (not illustrated). In this aspect, the design

service(s) **126** selects the second design template based on the keywords **127** (updated keywords) in response to receiving a selection of the reset option. Alternatively, the design service(s) **126**, subsequent to generating the design **103**, refrains from updating the design template if the reset option is not selected. In this aspect, the user **104** can update the user input **109** to incrementally update the design **103** by updating the design content **119** (without changing the design template), as described herein. In a particular aspect, the GUI **181** does not include the reset option and the design service(s) **126** selects the second design template based on the keywords **127** independently of receiving a selection of any reset option.

(51) In a particular aspect, the design service(s) **126** (e.g., in response to receiving a selection of the reset option) selects a second design template based on the keywords **127** (e.g., “red”, “invitation”, “red invitation”, “puppy”, “balloons”). The design service(s) **126** generates a second design based on the second design template. For example, the design service(s) **126** adds a second design element to the second design in response to determining that the second design template includes a first design element. The design service(s) **126** copies one or more attributes of the first design element to the second design element.

(52) In a particular aspect, generating the second design includes applying the design **103** to the second design template. For example, the design service(s) **126** generates a content signature of the design **103**. The content signature indicates attributes of the design element(s) **121** of the design **103**. The design service(s) **126** applies the content signature to the second design, as described herein. The design service(s) **126** generates element match data indicating elements of the design **103** that match elements of the second design. For example, the design service(s) **126** generates a first list indicating the design element(s) **121** of the design **103** and a second list indicating design elements of the second design (copied from the second design template). The design service(s) **126**, in response to determining that the design element **123** of the first list has the same content role and the same page tag as a second design element of the second list, updates element match data to indicate that the design element **123** matches the second design element of the second list. For example, a title element on page 1 of the design **103** matches a title element on page 1 of the second design template. Design elements that are added to the element match data are removed from the first list and the second list. For example, the design service(s) **126** removes the design element **123** from the first list, and removes the second design element from the second list.

(53) The design service(s) **126** transfers content from the design element(s) **121** to matching design elements of the second design. In a particular aspect, the design service(s) **126**, in response to determining that the element match data indicates that the design element **123** of the design **103** matches the second design element of the second design, copies the text information **173**, the media identifier **189**, the fill information **177**, or a combination thereof, of the design element **123** to the second design element. The second design element of the second design is formatted based on first format information of the second design template, second format information indicated by the text information **173**, or both.

(54) In a particular aspect, the design service(s) **126** selects second design content (e.g., an image of balloons) based on the keywords **127**, updates the second design based on the second design content, and updates the GUI **181** based on a design image **829** of the second design, as described with reference to FIG. 4.

(55) Although examples described herein include the keywords **127** and text elements in the English language, it should be understood that in some examples the design service(s) **126** support internationalization. In a particular example, the design service(s) **126** selects the design template **105** based at least in part on determining that the design template **105** has a language tag that matches a language (e.g., Arabic) that is indicated by the search query **117** (e.g., A card that says “Happy Birthday” in Arabic). In another example, the design service(s) **126** selects the design template **105** (e.g., a birthday card template) independently of a language tag and generates the design **103** based on the design template **105**. In this example, the design service(s) **126** performs

language translation on text elements of the design **103** from a first language (e.g., English) associated with the design template **105** to a second language (e.g., Arabic) indicated by the search query **117**.

(56) A design image (e.g., the design image **129**, the design image **729**, or the design image **829**) is selectable to edit a corresponding design. For example, the design service(s) **126**, in response to receiving a selection of the design image **129**, updates the GUI **181** (e.g., generates a second GUI) to include one or more options to edit the corresponding design (e.g., the design **103**), a save option to store the corresponding design, or a combination thereof. The GUI **181** may include a display portion that displays the design **103**. In various aspects, the GUI **181** may enable at least any graphic design function that is generally found in a standalone graphic design application, although it is to be appreciated that the GUI **181** may enable other functionality as well. The design service(s) **126** provides the updated version of the GUI **181** to the client device **192**. In the example illustrated in FIG. **9**, the user **104** updates a text element of the design **103** to replace default text (e.g., “The Garden Ballroom”) applied from the design template **105** with customized text (e.g., “Bow Wow Park”).

(57) In a particular aspect, the user **104** selects the options to edit the design **103**. The client device **192**, in response to receiving the user input **109** indicating the edits to the design **103**, sends the user input **109** to the computing environment. The design service(s) **126** updates the design **103** based on the user input **109**. In a particular aspect, the design service(s) **126**, in response to receiving a selection of the save option, stores the updated version of the design **103** in the design database(s) **184**. In a particular aspect, the design service(s) **126** updates, at a first time, the search selection data **107** to indicate that the design **103** is saved at the first time by the user **104** and that the design **103** is based on the design template **105** and the search query **117**. The system **100** thus enables automatic generation of the design **103** based on the search query **117** and updates to the search query **117**. The design **103** includes a pre-applied applied layout based on the design template **105** (selected based on the keywords **127**). The design **103** is updated based on design content **119** that is selected based on the keywords **127**. The design **103** is thus customized based on the search query **117**, and the system **100** thus provides a rich and robust way to dynamically create and edit graphical/visual designs based on natural language text entry.

(58) Whereas FIG. **1** illustrates an example of a system **100** that is operable to generate a design based on a user search query (examples of which are illustrated and described with reference to FIGS. **2-7**), it is to be understood that operation of such a system may include various local and/or cloud-based components that are not illustrated in FIG. **1**. For example, FIG. **10** depicts, in greater detail, a logical diagram of a system **1000** operable to support aspects of the present disclosure. Although illustrated in a logical diagram, it is to be understood that the various components of the system **1000** may include or may correspond to computer hardware, computer software, or a combination thereof. Moreover, components of the system **1000** may be configured to communicate with each other, for example via wired network(s) and/or wireless network(s). Thus, in various implementations, operations described herein as being performed by a particular component of FIG. **10** may be performed by dedicated hardware, software, or a combination thereof corresponding to the particular component.

(59) In particular aspects, the system **1000** supports browser-based and mobile application-based access to a graphic design website. To illustrate, the graphic design website may correspond to the GUI **181** of FIG. **1**, and some or all of the components in the system **1000** may be implemented by the computing environment, one or more servers, the design database(s) **184** of FIG. **1**, a document database **1084**, a media database **1085**, the client device **192** of FIG. **1**, one or more message queues **1042**, load balancers **1006**, and/or an export database **1088**. For example, users may log in to the graphic design website and create, edit, and save graphic designs. The graphic design website may support operations including, but not limited to, creating a new design based on a search query (as described with reference to the preceding figures), applying a template to a design, searching

for and adding images to a design, saving a design, publishing of a design, etc.

(60) The system **1000** includes a web front end **1002** executed at a platform as a service (PaaS) provider **1004**. The PaaS provider **1004** may enable at-scale deployment of software as a service (SaaS) applications, such as web applications. For example, the web front end **1002** may represent one, five, ten, or some other number of instances of a SaaS application that executes on hardware owned by and/or leased from the PaaS provider **1004**. Each such instance of the front end SaaS application may be accessible via the Internet. In some examples, a browser or a mobile application executed by a user's computing device may access the web front end **1002** via the CDN **101**. The CDN **101** may also be configured to cache static content (e.g., thumbnails, static images, static web content, etc.). To illustrate, when a user requests a particular content item, such as a particular image, if the CDN **101** (e.g., a server thereof) stores a copy of the image, the CDN may respond to the request without passing the request further into the service infrastructure shown in FIG. **10**. The CDN **101** may thus include multiple geographically distributed “edge” servers that cache content.

(61) The web front end **1002** may be configured to serve a graphic design website to requestors, such as by responding to hypertext transfer protocol (HTTP) GET requests, HTTP POST requests, etc. The web front end **1002** may also be configured to respond to application programming interface (API) calls originating from web browsers and/or mobile apps. In some examples, the web front end **1002** may utilize an in-memory cache **1003** that stores session data. To illustrate, when a user logs in to the graphic design website served by the web front end **1002**, information regarding the user (e.g., display name, photo, subscription level, e-mail address, etc.) may be stored in the in-memory cache **1003** for rapid retrieval, processing, and/or rendering on some or all of the webpages of the graphic design website. Use of the in-memory cache **1003** may thus enable quickly serving webpages without accessing backend systems to determine user information.

(62) The system **1000** includes the load balancers **1006** configured to support communication and workload distribution from the web front end **1002** to a plurality of backend services or microservices that execute, for example, on one or more virtual machines in a cloud computing environment. Each such virtual machine may be implemented using physical hardware that is owned by and/or leased by a cloud service provider. The system **1000** may support scaled, dynamic deployment of services based on scale groupings in a multi-tiered configuration. In particular aspects, services may be grouped in certain virtual machines based on expected load patterns for the services. The architecture of the system **1000** may enable the reorganization of these groupings with low effort, based on changes in observed load patterns or the introduction of new services. Separating services across multiple virtual machines may also enable scaling specific service groups in response to increases in certain kinds of loads. This scaling may occur automatically based on monitoring of load, and may also be triggered manually in anticipation of load increases. To illustrate, if an increasing number of export tasks (e.g., publish tasks) are observed (e.g., based on percentage of busy vs. idle time), more virtual machines that include instances of the export service **1010** and/or an export worker **1048** may dynamically be instantiated to service the demand.

(63) Conversely, if an administrator of the system **1000** anticipates export volume (e.g., publish volume) to increase, the administrator may manually initialize additional virtual machines that include instances of the export service **1010** and/or the export worker **1048**. When export volume decreases, the virtual machines may be automatically (e.g., based on percentage of busy vs. idle time) or manually scaled down.

(64) In the example of FIG. **10**, a first tier of services includes synchronous “request-response” remote procedure call (RPC) services and a second tier includes asynchronous “worker” services. A synchronous RPC service may receive first data from a requestor, perform one or more processing functions, and return second data to the requestor. Such synchronous request-response services may be used for tasks that can be completed relatively quickly, e.g., in three seconds or less, because in some examples a requestor (e.g., a web browser or mobile app) may block while waiting for a response. Conversely, the asynchronous worker services of the second tier may be used for

background tasks and other tasks that cannot or may not be completed relatively quickly.

(65) In FIG. 10, the first tier includes a first scale group **1022** (designated “A”), a second scale group **1040** (designated “G”), and a third scale group **1041** (designated “E”). The first scale group **1022** includes a billing RPC service **1008**, an export service **1010**, an import RPC service **1012**, a license RPC service **1014**, a media service **1016**, a review RPC service **1018**, and a share RPC service **1020**. To illustrate, a virtual machine corresponding to the first scale group **1022** may execute the illustrated RPC services.

(66) The billing RPC service **1008** may be configured to perform synchronous billing tasks (e.g., checking account balance). The export service **1010** may be used to perform synchronous graphic design export tasks (e.g., publishing a design). The import RPC service **1012** may be configured to enable user importing of media assets for use in graphic design creation/editing. The license RPC service **1014** may be configured to enforce image licenses (e.g., one-time-use licenses). The media service **1016** may be configured to track and catalog media assets available via the system **1000**, including but not limited to user-uploaded and/or library images, templates, fonts, etc. The review RPC service **1018** may be configured to facilitate acceptance of contributor media (e.g., uploading of media assets by authorized contributors of the graphic design website). The share RPC service **1020** may enable sharing of graphic designs with other users, such as via social media accounts.

(67) The second scale group **1040** includes a comment RPC service **1024**, a document service **1026**, a folder RPC service **1028**, a notification RPC service **1030**, a print RPC service **1032**, a profile RPC service **1034**, a search RPC service **1036**, and a social RPC service **1038**. The comment RPC service **1024** may enable users to comment on graphic designs in comment threads. The document service **1026** may enable core design creation, design updating, and design deletion functionality of the graphic design website. The folder RPC service **1028** may enable navigating into and out of different folders associated with a user's account. The notification RPC service **1030** may enable generating and serving notifications to users via the graphic design website (e.g., when a user's design is “liked” or commented on). In some aspects, multiple such notifications may be coalesced into an e-mail that is sent to an e-mail address of the user or that is shown on a pop-up when the user accesses the graphic design website. The print RPC service **1032** may be a format service that enables sending a graphic design to a third-party printer, for example to be printed and then delivered to a physical mailing address of the user. The profile RPC service **1034** may enable users to manage their graphic design website account. The search RPC service **1036** may enable searching for images, templates, designs, etc. The social RPC service **1038** may enable users to perform social networking engagement on the graphic design website (e.g., “follow” another user, “like” another user's design, etc.) and may generate “feeds” of designs created by followed users. The third scale group **1041** includes a compute RPC service **1043** that may be dedicated for performing certain fast high-CPU-utilization operations, such as hashing and solving of passwords.

(68) The second tier of services in FIG. 10 includes a fourth scale group **1058** (designated “D”), a fifth scale group **1064** (designated “F”), a sixth scale group **1068** (designated “C”), a seventh scale group **1076** (designated “H”), and an eighth scale group **1072** (designated “B”). The fourth scale group **1058** includes a billing worker **1044**, a document worker **1046**, an export worker **1048**, a folder worker **1050**, a search worker **1052**, a share worker **1054**, and a social worker **1056**. The fifth scale group **1064** includes a low priority import worker **1060** and a review worker **1062**. The sixth scale group **1068** includes a high priority import worker **1066**. The seventh scale group **1076** includes a low priority export worker **1074**. The eighth scale group **1072** includes a medium/high priority export worker **1070**. In illustrative aspects, the asynchronous worker services in FIG. 10 may perform longer and/or more complex versions of the operations performed by the corresponding RPC services of FIG. 10. In a particular implementation, the priority designations shown in FIG. 10 (e.g., “low,” “medium,” and “high”) correspond to an amount of dedicated processing resources. “High” priority may be used for jobs that are triggered by user actions whereas “medium” or “low” priority may be used for jobs that are not triggered by user actions

(e.g., background jobs).

(69) It is to be understood that the various services and groupings thereof shown in FIG. **10** is for illustration only and is not to be considered limiting. In alternative aspects, more, fewer, and/or different services may be present in the system. Moreover, a different grouping of services into different scale groups than shown in FIG. **10** may be used. In some aspects, each of the scale groups shown in FIG. **10** corresponds to a virtual hardware instance, i.e., a virtual machine running at a cloud services provider. Thus, at any given time, there may be one or more active/executing instances of each of the scale groups, and the specific numbers of active/executing virtual machine instances may dynamically change based on the overall workload being managed by the load balancers **1006**. Coordination and configuration of the various instances, including communication between instances and/or services executed therein, may be managed by a coordination tool **1005**, which may itself be a cloud-based system.

(70) During operation, the synchronous RPC services of the first tier may be configured to communicate with the asynchronous workers' services of the second tier via the message queues **1042** and may utilize the asynchronous workers to perform time-consuming tasks. For example, the export service **1010** may receive a request that a user wants to export a graphical design. The export service **1010** may push a corresponding work request into the message queues **1042**. An export worker (e.g., the medium/high priority export worker **1070**) may respond to the work request by rendering the graphical design to a web output. The web output may then be passed back to the requesting user. As another example, the folder RPC service **1028** may receive a request indicating that a user wants to share a folder of graphic designs with another user. The folder RPC service **1028** may use the message queues **1042** to request that the folder worker **1050** set the corresponding permissions on the folder and on each of the items in that folder. As yet another example, asynchronous workers may be used to perform routine background tasks in the system **1000**, such as daily verification of subscription levels.

(71) During certain operations at the system **1000**, the illustrated services may access one or more databases or data stores. For example, the document database **1084** may store files corresponding to user-created graphic designs and the media database **1085** may store image uploads, fonts, and templates that are accessible via the graphic design website. In an illustrative aspect, the document database **1084** may be accessed by the document worker **1046** when a user opens or saves a graphic design. In another illustrative aspect, the media database **1085** may be accessed by the media service **1016** when a user uploads or retrieves an image, a template, or a font.

(72) The system **1000** may also include one or more of a license database **1086**, a share database **1087**, the export database **1088**, a profile database **1089**, or a social database **1090**. The license database **1086** may be used by the license RPC service **1014** to track licenses (e.g., one-time-use licenses for images) that have been acquired by a user. The share database **1087** may be used by the share RPC service **1020** or the share worker **1054** to manage sharing of graphic designs with other users (e.g., via social network(s)). The export database **1088** may be used by the export service **1010**, the export worker **1048**, the low priority export worker **1074**, or the medium/high priority export worker **1070** to track user exports of their graphic designs into different output formats. The profile database **1089** may support searching for user profiles based on different criteria. The social database **1090** may support social media aspects of the graphic design website.

(73) The system **1000** may also include databases or data stores that support search functionality.

(74) For example, a media search database **1091** may be accessed by the search RPC service **1036** or the search worker **1052** when a user searches for images, templates, or fonts. As another example, a profile search database **1092** may be accessed by the profile RPC service **1034** when a user edits the profile information associated with their account. As yet another example, a document search database **1093** may be accessed by the document worker **1046** when a user searches for a previously saved graphic design.

(75) Certain operations at the system **1000** may involve accessing “external” services that are not

part of the core service oriented platform of the graphic design website. Examples of such external services may include, but are not limited to, an analytics service **1078**, a metrics service **1079**, a billing service **1080**, an e-mail service **1081**, an open authorization (OAuth) service **1082**, or a logging service **1083**. The analytics service **1078** may receive analytics events (e.g., messages) each time a user performs an action on the graphic design website, and may enable per-user and macro level analytics of graphic design website workflow, operations, and performance. The metrics service **1079** may collect and provide performance information regarding the various components of the system **1000**. The billing service **1080** may interface to one or more external payment providers, such as for credit card processing, mobile payment processing, etc. The e-mail service **1081** may enable generating and/or sending e-mails to users (e.g., for notifications, password reset, etc.). The OAuth service **1082** may enable federated logins to the graphic design website using social network credentials and may also support authentication with certain publish endpoints. The logging service **1083** may index events/messages that are generated by the components of the system **1000** for later diagnostic searching.

(76) One example of an operation that may be supported by the system **1000** is logging in to the graphic design website via an Internet browser or a mobile app. A login request may be received by the web front end **1002** from the Internet browser or mobile app. The login request may be routed by the load balancers **1006** to an instance of the profile RPC service **1034**, which may “look up” the user in the profile database **1089** and may authenticate the user (federated social networking login may involve the OAuth service **1082**). Once the user is authenticated, the graphic design website may load a custom homepage for the user, which may include operations being performed by an instance of the folder RPC service **1028**, the document worker **1046**, etc. The custom homepage may then be returned to the user's Internet browser or mobile app.

(77) Another example of an operation that may be supported by the system **1000** is starting a new design or opening an existing design. When a user clicks on a new design template, the web front end **1002** may provide the user's Internet browser or mobile application a graphical design interface (e.g., HTML code) that is executable to edit the new or existing design. In the case of loading an existing design, the document service **1026** and/or the document worker **1046** may access the design database(s) **184** to retrieve the existing design, and the media service **1016** may load media assets that are included in the design from the media database **1085**.

(78) Another example of an operation that may be supported by the system **1000** is when a user updates a design using the graphical design interface served by the graphic design website. As the user updates the design, the updates may be processed by the document service **1026** and/or the document worker **1046**. For example, the service(s) may process image or text movement, insertion, deletion, resizing, recoloring, etc.

(79) Another example of an operation that may be supported by the system **1000** is when a user imports an image. In this case, the import RPC service **1012** and/or one of the import workers **1060**, **1066** may receive a user-uploaded image, process the image, and integrate the image into the media database **1085** for subsequent retrieval by the media service **1016**.

(80) Another example of an operation that may be supported by the system **1000** is when a user applies a template to a design. In this case, the media search database **1091** may provide via the media service **1016**, media assets corresponding to the template. The insertion of the media assets into the design may then be processed by the document service **1026** and/or the document worker **1046** as an update to the user's design.

(81) Another example of an operation that may be supported by the system **1000** is when a user searches for a media asset and then drags-and-drops the media asset into a design. For example, when a user enters a search query “dog,” the search RPC service **1036** and/or the search worker **1052** may access the media search database **1091** to retrieve search results for “dog.” The search results may be presented by the user's Internet browser or mobile app. When the user adds a particular search result, e.g., a particular image of a dog, to their design, the document service **1026**

or the document worker **1046** may process the corresponding update to the design.

(82) In a particular aspect, the system **1000** may support generating a design based on a search query. For example, a user may generate a design by providing a search query, as described with reference to the preceding FIGS. In some aspects, component(s) of the system **1000** (and/or the system **100**) may perform a method **1100** of FIG. **11** to generate and/or update the design based on the search query and/or modifications to the search query.

(83) The method **1100** includes receiving first user input indicating a first search query, at **1102**. For example, the design service(s) **126** of FIG. **1** receives the user input **109** via a network interface from the CDN **101**. The user input **109** indicates the search query **117**, as described with reference to FIGS. **1-6**.

(84) The method **1100** also includes identifying one or more first keywords by parsing the first search query, at **1104**. For example, the design service(s) **126** of FIG. **1** identifies the keywords **127** (e.g., “red”, “invitation”, and “puppy”) by parsing the search query **117** (e.g., “A red invitation with a puppy”), as described with reference to FIGS. **1-6**.

(85) The method **1100** further includes selecting a first design template based on the one or more first keywords, at **1106**. For example, the design service(s) **126** of FIG. **1** selects the design template **105** (e.g., a birthday template with a red color tag) based on the keywords **127**, as described with reference to FIGS. **1** and **4**.

(86) The method **1100** also includes selecting first design content based on the one or more first keywords, at **1108**. For example, the design service(s) **126** of FIG. **1** selects the design content **119** (e.g., an image of a dog) based on the keywords **127** (e.g., “puppy”), as described with reference to FIGS. **4-6**. The design content **119** includes at least one of the font **125**, the fill **135**, or the image **115**.

(87) The method **1100** further includes generating a first design based on the first design template, at **1110**. For example, the design service(s) **126** generates the design **103** based on the design template **105**, as described with reference to FIGS. **1** and **4**. To illustrate, generating the design **103** includes, based on determining that the design template **105** includes the design element **143** having the content role **179**, adding the design element **123** (e.g., an image element with a cupcake image) having the content role **179** to the design **103**.

(88) The method **1100** also includes updating the first design based on the first design content, at **1112**. For example, the design service(s) **126** of FIG. **1** updates the design **103** based on the design content **119** (e.g., the cupcake image in the image element is replaced with the image of the dog), as further described with reference to FIGS. **1** and **4-6**.

(89) The method **1100** further includes generating a first graphical user interface (GUI) including a first design image of the first design, at **1114**. For example, the design service(s) **126** of FIG. **1** generates the GUI **181** including the design image **129** of the design **103**, as described with reference to FIGS. **1** and **4-6**.

(90) The method **1100** also includes providing the first GUI to a device, at **1116**. For example, the design service(s) **126** of FIG. **1** provide the GUI **181**, via a network interface, to the client device **192**.

(91) The method **1100** thus enables automatically generating a design (e.g., the design **103**) based on a search query (e.g., the search query **117**) and updates to the search query. The keywords **127** are identified by parsing the search query **117**. The design template **105** is selected based on one or more of the keywords **127**. Design content **119** is selected based on one or more of the keywords **127**. The design **103** is generated based on the design template **105**. The design **103** is updated based on the design content **119**. The design **103** is thus customized based on the keywords **127**, and the method **1100** thus provides a rich and robust way to dynamically create and edit graphical/visual designs based on natural language text entry.

(92) Methods and devices that may implement aspect(s) of the various features of the present disclosure have been described with reference to the drawings. The drawings and the associated

descriptions are provided to illustrate aspects and not to limit the scope of the present disclosure. Reference to any specific aspect, aspect, or implementation is intended to indicate that a particular feature, structure, or characteristic described in connection therewith is included in at least an aspect, aspect, or implementation of the present disclosure. However, the various uses of the terms “aspect,” “aspect,” or “implementation” are not necessarily all referring to the same single aspect, aspect, or implementation.

(93) In a particular aspect, an apparatus includes network interface and a processor. The network interface is configured to receive first user input from a client device. The first user input indicates a first search query. The processor is configured to identify one or more first keywords by parsing the first search query. The processor is also configured to select a first design template based on the one or more first keywords. The processor is further configured to select first design content based on the one or more first keywords. The first design content includes at least one of a first font, a first fill, or a first image. The processor is also configured to generate a first design based on the first design template. Generating the first design includes, based on determining that the first design template includes a first design element having a first content role, adding a second design element having the first content role to the first design. The processor is further configured to update the first design based on the first design content. The processor is also configured to generate a first graphical user interface (GUI) including a first design image of the first design. The processor is further configured to provide the first GUI via the network interface to the client device.

(94) In another particular aspect, a method includes receiving, at a first device from a second device, first user input indicating a first search query. The method also includes identifying, at the first device, one or more first keywords by parsing the first search query. The method further includes selecting, at the first device, a first design template based on the one or more first keywords. The method also includes selecting, at the first device, first design content based on the one or more first keywords. The first design content includes at least one of a first font, a first fill, or a first image. The method further includes generating, at the first device, a first design based on the first design template. Generating the first design includes, based on determining that the first design template includes a first design element having a first content role, adding a second design element having the first content role to the first design. The method also includes updating, at the first device, the first design based on the first design content. The method further includes generating, at the first device, a first graphical user interface (GUI) including a first design image of the first design. The method also includes providing the first GUI from the first device to the second device.

(95) In another particular aspect, a computer-readable storage device stores instructions that, when executed by a processor, cause the processor to perform operations including receiving first user input indicating a first search query. The operations also include identifying one or more first keywords by parsing the first search query. The operations further include selecting a first design template based on the one or more first keywords. The operations also include selecting first design content based on the one or more first keywords. The first design content includes at least one of a first font, a first fill, or a first image. The operations further include generating a first design based on the first design template. Generating the first design includes, based on determining that the first design template includes a first design element having a first content role, adding a second design element having the first content role to the first design. The operations also include updating the first design based on the first design content. The operations further include generating a first graphical user interface (GUI) including a first image of the first design. The operations also include providing the first GUI to a client device.

(96) In the foregoing description, specific details are given to provide a thorough understanding of the present disclosure. However, it will be understood by one of ordinary skill in the art that the present disclosure can be practiced without these specific details. Well-known structures and

techniques may not be shown in detail, in order to avoid obscuring the illustrated aspects. For example, algorithms may be shown in block diagrams.

(97) It is noted that aspects may be described as a process that is depicted as a flowchart, a flow diagram, a structure diagram, or a block diagram. Although a flowchart may describe operations as a sequential process, many of the operations can be performed in parallel or at least partially concurrently. In addition, the order of the operations can be rearranged. A process is terminated when its operations are completed. A process can correspond to a method, a function, a procedure, a subroutine, a subprogram, etc. When a process corresponds to a function, its termination corresponds to a return of the function to the calling function or the main function.

(98) Moreover, storage may represent one or more devices that store data, including but not limited to random access memory (RAM), flash memory, read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), registers, a hard disk, a removable disk, a disc-based memory (e.g., compact disc read-only memory (CD-ROM)), a flash memory device, and/or other computer-readable or processor-readable media to store information. As used herein, a computer-readable or processor-readable storage medium/device is not a signal.

(99) Furthermore, aspects can be implemented by hardware, software, firmware, middleware, microcode, or any combination thereof. When implemented in software, firmware, middleware or microcode, program code or code segments to perform the tasks can be stored in a memory or other storage. One or more than one processor can perform tasks in series, distributed, concurrently, or in parallel. In some examples, a virtual computer system can be constructed to implement one or more of the methods or functionality as described herein. A code segment can represent a procedure, a function, a subprogram, a program, a routine, a subroutine, a module, a software package, a class, or a combination of instructions, data structures, or program statements. A code segment can be coupled to another code segment or a hardware circuit by passing and/or receiving information, data, arguments, parameters, or memory contents. Information, arguments, parameters, data, etc. can be passed, forwarded, or transmitted through a suitable means including memory sharing, message passing, token passing, network transmission, etc.

(100) Although the present disclosure has been described with a degree of particularity, it is understood that the present disclosure has been made by way of example and that other versions are possible. As various changes could be made in the above description without departing from the scope of the disclosure, it is intended that all matter contained in the above description or shown in the accompanying drawings shall be illustrative and not used in a limiting sense.

(101) All features disclosed in the specification, including the claims, abstract, and drawings, and all the steps in any method or process disclosed, can be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. Each feature disclosed in the specification, including the claims, abstract, and drawings, can be replaced by alternative features serving the same, equivalent or similar purpose, unless expressly stated otherwise.

(102) Thus, unless expressly stated otherwise, each feature disclosed is one example only of a generic series of equivalent or similar features. The above-disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover all such modifications, enhancements, and other aspects, which fall within the true scope of the present disclosure. Thus, to the maximum extent allowed by law, the scope of the present disclosure is to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing description.

Claims

1. An apparatus comprising: a network interface configured to receive a first user input from a client device and a second user input, the first user input being text input defining a first search query and the second user input indicating a second search query; and a processor configured to: identify one or more first keywords by parsing the first search query; after identifying the one or more first keywords, select a first design template based on specific keywords of the one or more first keywords; after identifying the one or more first keywords, select first design content based on the specific keywords of the one or more first keywords, the first design content including at least one first image; after selecting first design template, generate a first design based on the first design template, wherein generating the first design includes, based on determining that the first design template includes a first design element having a first content role, adding a second design element having the first content role to the first design; after generating the first design and before presenting the first design to the client device, generate a first updated version of the first design based on the first design content; after generating the first updated version of the first design, generate a first graphical user interface (GUI) including a first design image of the first updated version of the first design; after generating the first GUI, provide the first GUI via the network interface to the client device; identify one or more second keywords by parsing the second search query; select a second design template based on the one or more second keywords; generate a second updated version of the first design by applying the second design template to the first design; generate a second GUI that includes a second design image of the second updated version of the first design; and provide the second GUI to the client device, wherein generating the second updated version of the first design by applying the second design template to the first design includes: generating a content signature of the first design, the content signature indicating that the second design element of the first design has the first content role; generating a second design based on the second design template, wherein generating the second design includes, based on determining that the second design template includes a third design element having the first content role, adding a fourth design element having the first content role to the second design; updating the second design by applying the content signature to the second design, wherein applying the content signature to the second design includes transferring content from the second design element to the fourth design element; and designating the second design as the second updated version of the first design.

2. The apparatus of claim 1, wherein the network interface is further configured to receive a selection of the first design image, wherein the processor is further configured to: in response to receiving the selection of the first design image, generate a third GUI that includes one or more options to edit the first updated version of the first design; and provide the third GUI via the network interface to the client device.

3. The apparatus of claim 2, wherein the processor is further configured to, in response to receiving a selection of a save option of the third GUI, update search selection data to indicate that the first updated version of the first design is saved by a user and that the first updated version of the first design is based on the first search query, the first design template, and the first design content.

4. The apparatus of claim 1, wherein the processor is further configured to select the first design template based at least in part on determining that a tag of the first design template matches at least one of the one or more first keywords.

5. The apparatus of claim 1, wherein the processor is further configured to select the first design content based at least in part on determining that a tag of the first design content matches at least one of the one or more first keywords.

6. The apparatus of claim 1, wherein the processor is further configured to: identify a first plurality of design templates based on the one or more first keywords; and determine scores of the first plurality of design templates, the scores including a first score of the first design template, wherein the first design template is selected based at least in part on determining that the first score satisfies

a score threshold.

7. The apparatus of claim 6, wherein the processor is further configured to determine the first score based on a count of the one or more first keywords that match at least one tag of the first design template.

8. The apparatus of claim 6, wherein the processor is further configured to: retrieve search selection data indicating that a third design was saved by a user at a first time and that the third design is based on a third search query and the first design template; and determine, at a second time, the first score based on a comparison of the first time and the second time, a comparison of the first search query and the third search query, a comparison of first user characteristics of the user and second user characteristics of a logged-in user, or a combination thereof.

9. The apparatus of claim 1, wherein the processor is further configured to: select a third design template based on the one or more first keywords; select second design content based on the one or more first keywords; generate a third design based on the third design template; and generate an updated version of the third design based on the second design content, wherein the first GUI includes a third design image of the updated version of the third design.

10. The apparatus of claim 1, wherein the network interface is further configured to receive third user input indicating a third search query, and wherein the processor is further configured to: identify one or more third keywords by parsing the third search query; select second design content based on the one or more third keywords; generate a third updated version of the first design based on the second design content; generate a third GUI that includes a third design image of the third updated version of the first design; and provide the third GUI to the client device.

11. The apparatus of claim 10, wherein the second design content includes at least one of a second font, a second fill, or a second image, and wherein generating the third updated version of the first design based on the second design content includes replacing at least one of a first font, a first fill, or the first image in the first design with a corresponding one of the second font, the second fill, or the second image.

12. A method comprising: receiving, at a first device from a second device, a first user input being text input defining a first search query and a second user input indicating a second search query; identifying, at the first device, one or more first keywords by parsing the first search query; after identifying the one or more first keywords, selecting, at the first device, a first design template based on specific keywords of the one or more first keywords; after identifying the one or more first keywords, selecting, at the first device, first design content based on the specific keywords of the one or more first keywords, the first design content including at least one first image; after selecting the first design template, generating, at the first device, a first design based on the first design template, wherein generating the first design includes, based on determining that the first design template includes a first design element having a first content role, adding a second design element having the first content role to the first design; after generating the first design and before presenting the first design to the second device, generating, at the first device, a first updated version of the first design based on the first design content; generating, at the first device, a first graphical user interface (GUI) including a first design image of the first updated version of the first design; providing the first GUI from the first device to the second device; identifying one or more second keywords by parsing the second search query; selecting a second design template based on the one or more second keywords; generating a second updated version of the first design by applying the second design template to the first design; generating a second GUI that includes a second design image of the second updated version of the first design; and providing the second GUI to the client device, wherein generating the second updated version of the first design by applying the second design template to the first design includes: generating a content signature of the first design, the content signature indicating that the second design element of the first design has the first content role; generating a second design based on the second design template, wherein the generating the second design includes, based on determining that the second design template

includes a third design element having the first content role, adding a fourth design element having the first content role to the second design; updating the second design by applying the content signature to the second design, wherein applying the content signature to the second design includes transferring content from the second design element to the fourth design element; and designating the second design as the second updated version of the first design.

13. The method of claim 12, further comprising: receiving, at the first device from the second device, a selection of the first design image; in response to receiving the selection of the first design image, generating a third GUI that includes one or more options to edit the first updated version of the first design; and providing the third GUI from the first device to the second device.

14. The method of claim 12, further comprising selecting the first design template based at least in part on determining that a tag of the first design template matches at least one of the one or more first keywords.

15. The method of claim 12, further comprising: identifying, at the first device, a first plurality of design templates based on the one or more first keywords; and determining scores of the first plurality of design templates, the scores including a first score of the first design template, wherein the first design template is selected based at least in part on determining that the first score satisfies a score threshold.

16. A computer-readable storage device storing instructions that, when executed by a processor cause the processor to perform operations comprising: receiving a first user input being text input defining a first search query and a second user input indicating a second search query; identifying one or more first keywords by parsing the first search query; after identifying the one or more first keywords, selecting a first design template based on specific keywords of the one or more first keywords; after identifying the one or more first keywords, selecting first design content based on the specific keywords of the one or more first keywords, the first design content including at least one first image; after selecting first design template, generating a first design based on the first design template, wherein generating the first design includes, based on determining that the first design template includes a first design element having a first content role, adding a second design element having the first content role to the first design; after generating the first design and before presenting the first design to a client device, generating a first updated version of the first design based on the first design content; after generating a first updated version of the first design, generating a first graphical user interface (GUI) including a first image of the first updated version of the first design; after generating the first GUI, providing the first GUI to a client device; identifying one or more second keywords by parsing the second search query; selecting a second design template based on the one or more second keywords; generating a second updated version of the first design by applying the second design template to the first design; generating a second GUI that includes a second design image of the second updated version of the first design; and providing the second GUI to the client device; wherein generating the second updated version of the first design by applying the second design template to the first design includes: generating a content signature of the first design, the content signature indicating that the second design element of the first design has the first content role; generating a second design based on the second design template, wherein the generating the second design includes, based on determining that the second design template includes a third design element having the first content role, adding a fourth design element having the first content role to the second design; updating the second design by applying the content signature to the second design, wherein applying the content signature to the second design includes transferring content from the second design element to the fourth design element; and designating the second design as the second updated version of the first design.

17. The computer-readable storage device of claim 16, wherein the operations further comprise: selecting a third design template based on the one or more first keywords; selecting second design content based on the one or more first keywords; generating a third design based on the third design template; and generating an updated version of the third design based on the second design

content, wherein the first GUI includes a third design image of the updated version of the third design.

18. The computer-readable storage device of claim 16, wherein the operations further comprise: receiving third user input indicating a third search query; identifying one or more third keywords by parsing the third search query; selecting second design content based on the one or more third keywords; generating a third updated version of the first design based on the second design content; generating a third GUI that includes a third design image of the third updated version of the first design; and providing the third GUI to the client device.
